Emergency dispatch and communication system performance

"From full-text studies, extract reported values with source and setting: ambulance crew time to accept a job; vehicle GPS/AVL update intervals; duplicate or unassigned dispatch rates during manual or degraded operation; multi-hop LoRa or Meshtastic delivery ratio and latency; CBBA, auction or quorum allocation performance under message loss. Note which values are missing." *Generated by LeapSpace*

Operational Values in Emergency Dispatch, Tracking, Mesh Communications, and Distributed Allocation at a Glance

Key Findings Table

Topic	Reported value(s)	Setting/source type	What is missing
Ambulance/job acceptance time	**Median notification acceptance 40 s** (IQR **27 s to 1 min 19 s**) [1]	Simulation-based feasibility study of ambulance/first-responder incident notification acceptance [1]	Routine field EMS studies directly measuring dispatch-to-crew acknowledgement/button-press/radio accept are largely absent [2] [3]
Adjacent EMS workflow intervals	Smartphone responder **turnout median 1:45 min** after alert acceptance [4]; dispatch-to-availability median **173 min** for transported vs **73 min** for non-transported patients [5]; offload delay median **11 min** [6]	Smartphone responder systems and EMS operational studies [4] [5] [6]	Refusal rates, full time-to-accept distributions, and consistent timestamp definitions often not reported [2] [4] [7] [8] [9]
GPS/AVL refresh intervals	**1 min** updates [10]; adaptive **200-300 ms**, **400-500 ms**, **800-1000 ms**, default **500 ms** [11]; **100 ms** updates with **8-12 ms** average latency [12]	Floating-car-data, smart navigation, digital-twin synchronization [10] [11] [12]	Emergency vehicle papers often omit exact polling interval or end-to-end latency [13] [14] [15] [16] [17]
Dispatch errors in degraded/manual settings	Failure to report in **7.3% of calls** [18]; **27.2%** of OHCA cases not correctly dispatched [19]; **5.5%** of non-emergent dispatches later treated by EMS [20]	Operational dispatch studies; fallback/manual dispatch literature [18] [19] [20]	No clean duplicate-dispatch rate or unassigned-job rate for manual/degraded ambulance dispatch was retrieved [21] [22] [23] [24]

Topic	Reported value(s)	Setting/source type	What is missing
Multi-hop LoRa delivery	Up to **94.02%** success from farthest node to gateway [25]; **99% PDR** in simple multi-hop [26]; **80% PDR** in 6-node urban flooding deployment [27]; **51%-77% PRR** in 4-hop flooding test [28]; **100%** receiving rate in underground network [29]	Implementations, field deployments, application networks [25] [26] [27] [28] [29]	Meshtastic field benchmarks with hop count, terrain, spacing, antenna, and latency details were not retrieved [30] [31]
LoRa latency/load effects	End-to-end delay **sub-400 ms** at lower spreading factors and **well above 1 s** at high spreading factors [32]; duty-cycle study DR fell **0.958→0.438** and **0.159→0.058** under higher constraint/load [33]; delivery in hospital mesh stayed **>99.5%** at **1 packet/hour** but fell to **~83.2%** at **26 packets/hour** [34]	LoRaWAN and mesh simulations/experiments [32] [33] [34]	Latency is much less consistently reported than delivery ratio [35] [36] [37]
CBBA/auction/quorum under message loss	CBBA reported conflict-free convergence and at least **50%** of Nash-equilibrium solutions within bounded timeframes [38] [39]; average sailing time **643.19 s** for CBBA vs **606.58 s** for DCBBA-TICC [40]; sequential bidding had **~13%** and **~19%** lower communication overhead than concurrent bidding [41]; Clustering-CBBA reduced communication steps **2745→592** (**78%**) [42]	Foundational theory, comparative simulation, clustered extensions [38] [39] [40] [41] [42]	Explicit loss thresholds tied to allocation success, missed-assignment rate, or quorum failure were not reported in the retrieved text [39] [43]

Direct Answer

The strongest directly reported values are in **AVL/GPS**, **LoRa**, and **distributed-allocation** studies, while the most operationally relevant **field EMS degraded-dispatch metrics** are often missing [10] [11] [12] [25] [26] [42].

For **ambulance crew time to accept a job**, the only direct value retrieved was a **simulation-based** median notification acceptance time of **40 s** (IQR **27 s to 1 min 19 s**) [1]. Field EMS papers more often report adjacent intervals such as turnout, dispatch-to-availability, or offload delay rather than dispatch-to-acknowledgement latency [4] [5] [6]. So the requested field value is mostly **missing** [2] [3].

For **vehicle GPS/AVL update intervals**, explicit values were available outside EMS: **1 min** refresh [10], adaptive **200-300 ms to 800-1000 ms** [11], and **100 ms** updates with **8-12 ms** latency [12]. But ambulance tracking papers commonly describe tracking capability without giving polling interval or latency [13] [14].

For **duplicate or unassigned dispatch during manual/degraded operation**, no retrieved full-text study gave a clean duplicate-assignment or unassigned-job rate for ambulance fallback/manual dispatch [21] [22] [23] [24]. Closest available values were **7.3%** failure-to-report calls [18], **27.2%** incorrect OHCA dispatch [19], and **5.5%** non-emergent dispatches later treated by EMS [20].

For **multi-hop LoRa**, reported delivery outcomes ranged from **51%-77%** PRR in a 4-hop flooding test [28] to **99% PDR** [26] and **100%** receiving rate in an underground network [29]. Latency was less commonly reported; one LoRaWAN study found **sub-400 ms** delay at lower spreading factors and **well above 1 s** at high spreading factors [32]. No retrieved **Meshtastic** benchmark provided the requested relay/path details [30] [31].

For **CBBA, auction, or quorum allocation under message loss**, the evidence supports robustness claims and communication-overhead comparisons, but not explicit packet-loss thresholds tied to allocation success [39] [43]. Quantified results include **643.19 s vs 606.58 s** sailing time for CBBA vs DCBBA-TICC [40], **~13%-19%** lower sequential-bidding overhead [41], and **78%** communication-step reduction by Clustering-CBBA [42].

Confidence level: High — the answer directly reflects the supplied extraction summary and repeatedly noted evidence gaps [2] [14] [24] [30] [39].

Study Scope

- **Time span covered by retrieved studies:** approximately 2005-2025 for EMS acceptance-time searches, with mixed historical and recent studies in AVL, LoRa, and distributed allocation [1] [27] [42] [44].
- **Disciplines/settings:** EMS operations, dispatch systems, vehicle tracking/AVL, LoRa/LoRaWAN networking, and decentralized task allocation [2] [25] [39] [45].
- **Methods represented:** simulations, field deployments, retrospective audits, operational studies, and theoretical/comparative algorithm studies [1] [12] [18] [34] [39].

Assumptions & Limitations

- Several requested metrics are available only through **proxy intervals** rather than direct operational measures, especially in EMS [1] [2] [4].

- Emergency-vehicle AVL studies often discuss tracking performance but omit **refresh frequency** and **end-to-end latency**, limiting direct comparison with non-EMS systems [13] [14] [17].
- Manual/degraded dispatch studies mostly provide **qualitative failure descriptions** rather than duplicate or unassigned rates [21] [22] [46].
- Meshtastic-specific peer-reviewed field benchmarks matching the requested metrics were **not retrieved** [30] [31].
- CBBA/auction/quorum papers generally report convergence or communication cost, but not explicit **message-loss thresholds** for success or failure [39] [43] [47].

Suggested Further Research

- Instrument field EMS systems to measure **dispatch notification to crew acknowledgement** directly, with standardized definitions [2] [3].
- Require ambulance AVL papers to report **polling interval, display freshness, and control-center latency** as standard operational metrics [13] [14].
- Quantify **duplicate dispatch, unassigned jobs, and reassignment delay** during CAD/radio outages and paper fallback [23] [24].
- Produce peer-reviewed **Meshtastic** field benchmarks including terrain, spacing, antenna setup, hop count, PDR, and end-to-end latency [30] [31].
- Run allocation experiments sweeping **message-loss rate, delay, and partition severity** with outputs such as conflict count, missed-task rate, and allocation success [43] [47].

Full Report

Foundational Context

The five requested metric families are not measured with the same granularity. EMS studies often center on clinically meaningful workflow intervals such as dispatch-to-arrival, turnaround, or offload delay, rather than low-level communications events such as “crew accepted job” [2] [48]. By contrast, networking and distributed-allocation studies more often expose exact technical quantities such as update period, packet delivery ratio, latency, or communication steps [12] [33] [42].

This difference matters for interpretation. A reported “acceptance time” may mean app acceptance, radio acknowledgement, turnout, or readiness for next case, and these are not interchangeable [2] [4] [49]. Similarly, “tracking update” may refer to GPS sampling, transmission cadence, control-center refresh, or delayed batch upload [44] [45]. In LoRa and task-allocation work, packet delivery and convergence under unreliable links are often measurable, but field emergency operations rarely log equivalent machine-readable failure events during degraded operation [22] [34] [47].

Confidence level: High — this framing is directly supported by the cross-domain synthesis and repeated reporting patterns in the extracted studies [2] [32] [45] [50] [51].

1. Ambulance Acceptance and Job Acknowledgement Timing

1.1 Directly reported acceptance-time values

Only one retrieved study directly reported a dispatch-notification acceptance interval analogous to "time to accept a job": a simulation-based feasibility trial measured **notification acceptance** with a **median of 40 s** and **IQR 27 s to 1 min 19 s** [1]. This is the clearest match to the user's requested quantity, but it is explicitly a **simulation study**, not routine field ambulance operations [1].

Other responder-alerting studies show that acceptance can be separated from later operational stages. One study of smartphone alerting measured alert acceptance followed by turnout and reported **median turnout time 1:45 min**, about one-third of total response time [4]. Another study documented **1,498 alerts** and **2,480 community first responder acceptances**, but also showed incomplete manual confirmation, with arrival documentation in only **69.2%** of cases [7]. A simulation-based lay first-responder dispatch study also measured initial notification to acceptance and acceptance-to-arrival as distinct phases, confirming that these steps are technically measurable even if often omitted in ambulance studies [8].

Taken together, these findings suggest the measurement problem is not conceptual difficulty but inconsistent operational logging in field EMS systems [1] [2].

1.2 Adjacent workflow intervals often reported instead

Because direct acceptance timing is scarce, adjacent intervals are the main available operational timings. A dispatch-to-availability metric showed median time from ambulance dispatch to readiness for the next case of **173 min** for transported patients and **73 min** for non-transported patients [5]. Offload delay, defined as ED arrival until crew report and stretcher handoff, had a **median of 11 min** in one pilot study [6]. Other EMS studies reported dispatch-to-arrival medians such as **7.2 min** (IQR **5.7-9.0**) and various turnaround substeps, but definitions varied substantially [48] [49] [52] [53].

These are useful operational measures, but they do not answer the narrower question of "how long did the crew take to accept the job?" [2] [48].

1.3 Missing values and comparability problems

The main missing value is a routine field measure of **dispatch-to-crew acknowledgement/acceptance latency** [2] [3]. Reporting compliance is also inconsistent: a CAD-based study found crew reporting compliance ranging from **52.8% to 94%** [54]. Reviews emphasize that agencies define interval boundaries differently, making cross-system comparison difficult [2] [55] [56].

Reported values extracted

- Notification acceptance median **40 s** (IQR **27 s to 1 min 19 s**) — simulation [1]
- Turnout median **1:45 min** after alert acceptance — smartphone responder study [4]

- Dispatch-to-availability median **173 min** transported, **73 min** non-transported [5]
- Offload delay median **11 min** [6]

Missing

- Field ambulance dispatch-to-accept values [2] [3]
- Refusal rates and full time-to-accept distributions in many alerting studies [4] [7] [8] [9]

Confidence level: Medium-High — the extracted values are clear, but the evidence base for the exact target metric is thin and relies partly on proxy measures [1] [2].

2. AVL and GPS Refresh Intervals in Vehicle Tracking

2.1 Explicit update intervals

The clearest reported AVL/GPS update cadences are from non-EMS tracking systems. One floating-car-data section-control system updated vehicle positions **every minute** [10]. A smart navigation system used a **default 500 ms** interval and adapted it to **200-300 ms** at high speed, **400-500 ms** at moderate speed, and **800-1000 ms** at low speed or while waiting [11]. A digital-twin synchronization setup updated every **100 ms** and reported average end-to-end update/prediction latency of **8-12 ms** per vehicle, explicitly characterizing performance as real-time [12].

A separate AVL system supported a **variable update rate** depending on user requirements, but the exact intervals were not disclosed in the supplied excerpt [57].

2.2 Communication-path dependence

Refresh cadence depends not only on GPS sampling but also on the transfer path. One AVL-based transit system described both **real-time radio transmission** and **nightly uploads** over automated high-speed links, showing that the same underlying data stream can be immediate or delayed depending on communications mode [44]. Another GPS-based transit system delayed transfers until WiFi coverage and sufficient stationary time were available, making data transfer opportunistic rather than continuous [58].

This directly supports the broader pattern that communications links, not just positioning hardware, constrain usable freshness [44] [45].

2.3 Emergency-vehicle tracking: capability described, cadence omitted

Ambulance-specific studies were less numerically explicit. A TETRA-based ambulance tracking system reported speed, direction, and location at preset intervals, but no exact update period was given; the paper also noted location inaccuracies likely tied to terminal age and transmission compression [13]. A GPS/web-map ambulance ETA study reported **72.8%** of arrivals predicted within **5 min** and mean absolute transport-time error of **3.5 min**, but did not report the AVL polling rate or end-to-end latency [14]. More general telematics papers similarly discussed capability and degraded-signal mitigation without giving refresh intervals [15] [16] [17].

Reported values extracted

- **1 min** update interval [10]
- **500 ms** default; **200-300 ms**, **400-500 ms**, **800-1000 ms** adaptive bands [11]
- **100 ms** updates with **8-12 ms** latency [12]
- Variable update rate supported, exact value not disclosed [57]

Missing

- Exact ambulance AVL polling interval [13] [14]
- Dispatcher display refresh and control-center latency in emergency settings [13] [17]

Confidence level: High — the non-EMS update values are explicit, and the ambulance omissions are clearly documented [10] [11] [14].

3. Duplicate or Unassigned Dispatch During Manual or Degraded Operation

3.1 Quantified operational errors

No retrieved study reported a clean **duplicate-dispatch rate** or **unassigned-job rate** for manual/degraded ambulance dispatch [21] [22] [24]. The nearest quantified figures are adjacent error indicators:

- **7.3%** average failure to report calls, partly driven by radio or dispatch acknowledgement difficulties [18]
- **27.2%** of OHCA cases not correctly dispatched [19]
- **5.5%** of non-emergent dispatches later treated by EMS [20]

Additional dispatch literature reports protocol sensitivity/specificity problems and time-of-day effects on missed priority assignment rather than duplicate or unassigned resource counts [59] [60].

3.2 What degraded/manual studies describe qualitatively

Fallback studies describe the right failure modes but seldom quantify them. A major-incident study reported that personnel on scene were not recorded through formal staging, preventing assignment to prioritized tasks and forcing repeated requests for outside resources; this is a direct example of **unassigned work**, but without a rate [21]. A systematic review found coordination problems leading to duplicated response efforts and ambulance unavailability, again without numerical duplicate-assignment rates [22]. Another paper described manual dispatch activation after communications-network failure and noted increased risk of verbal mistakes when phone-based or SCADA/EMS support was lost [23]. Dispatch-center failures due to operator, hardware, or software problems were also explicitly recognized [61].

3.3 Missingness is itself a major finding

Several studies state that operational outcomes are inconsistently defined or lack enough information for evaluation [50] [62]. One communication study noted that in **18%** of calls there was not enough information to assess whether a physician-manned unit should also have been dispatched [63]. Another prehospital communication paper found many arrivals unannounced or insufficiently described, but did not quantify duplicate assignment or reassignment delays [46].

Reported values extracted

- Failure to report: **7.3% of calls** [18]
- Incorrect OHCA dispatch: **27.2%** [19]
- Non-emergent dispatches later treated by EMS: **5.5%** [20]
- Insufficient information for physician-unit assessment: **18% of calls** [63]

Missing

- Duplicate-dispatch rate during manual fallback [22] [24]
- Unassigned-job rate during degraded operations [21] [23]
- Time-to-reassignment after dispatch failure [46]

Confidence level: Medium-High — the absence of the exact requested rates is strongly evidenced, but the available metrics are proxies rather than direct duplicate/unassigned counts [21] [24].

4. Multi-hop LoRa and Meshtastic Delivery Performance

4.1 Delivery ratio and reception success

The LoRa evidence base contains several directly reported delivery values:

- Up to **94.02%** success from the farthest node to the gateway in one multi-hop implementation [25]

- **99% PDR** in a simple multi-hop setup [26]
- **80% PDR** in a **6-node urban deployment** using LoRaBlink flooding [27]
- **51%-77% PRR** in a **4-hop relay-group** concurrent-transmission flooding test [28]
- **100% packet receiving rate** in an underground multi-hop soil-moisture network [29]

A custom network for earthquake early warning also reported improved PDR once the network exceeded three hops, though the excerpt did not include a numeric value [64].

4.2 Latency and load sensitivity

Latency values were scarcer than delivery figures. One LoRaWAN study reported end-to-end delay **below 400 ms** at lower spreading factors and **well above 1 s** at high spreading factors [32]. A hospital mesh simulation found delivery remained **above 99.5%** at **1 packet/hour** but dropped to **about 83.2%** at **26 packets/hour**; using **150-byte** packets caused roughly **2.1×** more collisions than **30-byte** packets [34]. Under duty-cycle enforcement in ns-3, delivery ratio fell from **0.958 to 0.438** for **N=25** under low load and from **0.159 to 0.058** under medium load [33].

These results show a consistent reliability-performance tradeoff: reachability can improve with relaying, but congestion, packet size, duty-cycle constraints, and PHY settings can sharply reduce performance [25] [33] [34].

4.3 Meshtastic-specific gap

No retrieved paper in the supplied corpus was a Meshtastic field test or benchmark reporting the requested **PDR, latency, hop count, terrain, node spacing, antenna setup, or per-hop versus end-to-end latency** [30] [31].

Reported values extracted

- **94.02%**, **99%**, **80%**, **51%-77%**, **100%** delivery/reception figures [25] [26] [27] [28] [29]
- Latency **<400 ms** to **>1 s** depending on spreading factor [32]
- Delivery **>99.5%** at **1 packet/hour**, **~83.2%** at **26 packets/hour** [34]
- Duty-cycle constrained DR **0.958→0.438** and **0.159→0.058** [33]

Missing

- Meshtastic benchmarks [30] [31]
- Hop-by-hop latency breakdown in many implementation papers [35] [36] [37]

Confidence level: High — multiple concrete values are available, and the Meshtastic evidence gap is explicit [25] [26] [30].

5. Task Allocation Performance Under Message Loss

5.1 Foundational CBBA/CBAA claims

Foundational sources describe CBBA/CBAA as decentralized task-allocation methods using local consensus over winning bids and prove convergence to a **conflict-free assignment** under reasonable scoring assumptions [39]. One source states CBBA can converge to at least **50% of Nash-equilibrium solutions** within bounded timeframes, albeit with substantial information exchange [38]. Additional papers describe robustness to communication-topology variation and inconsistencies in situational awareness, but do not provide explicit message-loss thresholds in the supplied text [39] [65] [66].

5.2 Effects of non-ideal communication

When communication is lossy, delayed, or asynchronous, the reported problems include **bid inconsistencies, consensus failures, and task conflicts** [47]. Later analyses also note that some earlier optimality guarantees may not always hold and that extra idle iterations can occur after convergence, motivating variants such as GCBBA and early convergence detection [67].

A dynamic underwater comparison gave one direct performance number: average sailing time was **643.19 s** for CBBA and **606.58 s** for DCBBA-TICC [40]. This reflects better execution efficiency for the variant, though not a direct packet-loss success rate [40].

5.3 Auction and clustered/quorum-style comparisons

Sequential bidding had lower communication overhead than concurrent/parallel bidding in homogeneous-channel allocation: about **13% lower** when varying users and about **19% lower** when varying channels [41]. In communication-degraded scenarios, one study compared Sequential Auction, Parallel Auction, Combinatorial Auction, G-Prim, and repeated Parallel Auction variants, but the supplied excerpt did not include explicit completion or failure percentages [43].

Clustering-CBBA preserved the same allocation as baseline CBBA without communication constraints while cutting communication steps from **2745** to **592**, a **78% reduction** [42]. Clustered and asynchronous variants are therefore promising for constrained links, though the provided excerpts do not quantify success probability under specific packet-loss rates [68] [69] [70].

Reported values extracted

- At least **50%** of Nash-equilibrium solutions within bounded timeframes [38]
- Average sailing time **643.19 s** CBBA vs **606.58 s** DCBBA-TICC [40]
- Sequential bidding communication overhead **~13%** and **~19%** lower [41]
- Communication steps **2745→592** (**78% reduction**) with Clustering-CBBA [42]

Missing

- Explicit packet-loss thresholds for convergence failure [39]
- Allocation success percentage under specified loss rates [43]
- Missed-assignment rate and quorum success/failure thresholds [43] [71]

Confidence level: Medium-High — the communication-overhead and comparative values are clear, but the exact "under message loss" success thresholds requested are mostly absent [43] [47].

Cross-Source Synthesis

Across the corpus, operational timing in EMS is often measured **indirectly**, whereas network and algorithm studies typically provide **precise technical metrics** [2] [12] [33] [42]. This creates an evidence imbalance: the most safety-critical field fallback scenarios are often the least numerically characterized [22] [46].

Taken together, these findings suggest three practical conclusions:

1. **Direct job-acceptance measurement is feasible but rarely institutionalized in field EMS** because studies can separate notification, acceptance, turnout, and arrival when the system captures those events [1].
2. **Communications-path design strongly shapes performance** across AVL, LoRa, and distributed allocation, affecting freshness, delivery ratio, delay, and convergence overhead [33] [44] [45] [47].
3. **Adaptive, asynchronous, or clustered designs appear to be the main mitigation strategy** under constrained links, whether by adaptive GPS updates, LoRa protocol design, or clustered CBBA [11] [42] [72].

Confidence level: High — these are cross-source inferences directly supported by repeated patterns in the extracted evidence [2] [34] [42] [45].

Conclusion

The requested operational values are available unevenly across domains. The most concrete extracted figures are:

- Ambulance/first-responder notification acceptance: **40 s median** in simulation [1]
- AVL/GPS updates: **100 ms** to **1 min**, with one real-time setup at **8-12 ms** latency [10] [12]
- Manual/degraded dispatch proxies: **7.3%**, **27.2%**, and **5.5%** adjacent error rates, but no direct duplicate/unassigned rates [18] [19] [20]
- Multi-hop LoRa delivery: roughly **51% to 100%** depending on topology and conditions, with latency from **sub-400 ms** to **>1 s** where reported [28] [29] [32]

- Allocation under constrained communication: **13%-19%** communication-overhead differences across auction styles and **78%** reduction via clustering, but no explicit loss threshold for allocation failure [41] [42]

The largest gaps are exactly the ones most relevant for degraded emergency operations: **field ambulance acknowledgement latency**, **duplicate/unassigned dispatch rates under fallback**, **Meshtastic field benchmarks**, and **allocation-success curves versus packet loss** [3] [24] [30] [43].

Overall confidence level: High — the report directly answers each requested metric family, distinguishes reported values from missing ones, and stays within the supplied evidence [1] [10] [18] [25] [42].

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